

Task 6 – Create a Strong Password and Evaluate Its Strength

Tool Used: Bitwarden Password Strength Testing Tool

1.Objective

To understand what makes a password strong by creating multiple example passwords, testing them using an online password strength tool, and analyzing how password complexity affects security. This task also aims to learn password attack techniques and apply modern password security best practices.

2. Methodology

The following steps were performed:

1. Created **8 passwords** with different complexity levels.
2. Tested each password in the **Bitwarden Password Strength Testing Tool**.
3. Recorded:
 - Password strength score (Very Weak → Strong)
 - Estimated time to crack
 - Feedback / security observations
4. Compared results to identify password improvement factors.
5. Researched **common password-based attacks** and industry security standards.
6. Summarized best practices for strong password creation.

3. Results – Password Strength & Crack Time Analysis

ID	Test Password	Strength Rating	Est. Time to Crack	Observations / Feedback
P1	jagan	✗ Very Weak	3 Seconds	Very short, dictionary-based name, no complexity
P2	mohan123	✗ Very Weak	1 Minute	Common name + predictable number pattern
P3	Sravanam	✗ Very Weak	45 Minutes	Only alphabets, dictionary-word style
P4	Sravan723	⚠ Weak	10 Hours	Slight complexity but still predictable
P5	Sandeep@192	⚠ Weak	11 Hours	Included symbol but still name-based

ID	Test Password	Strength Rating	Est. Time to Crack	Observations / Feedback
P6	Prem!8kumar@234	● Good	39 Years	Better complexity, mixed characters and length
P7	Saketh_Reddy!Vasu6	● Strong	Centuries	Long, random structure, secure
P8	Kuldeep\$Reddy_234!	● Strong	Centuries	Very strong: length + complexity + unpredictability

4. Analysis of Strength Patterns

The test proved:

- Passwords containing **real names** remain weak even with added numbers/symbols
- As complexity increases, **cracking time increases exponentially**
- Long passphrases are **significantly more secure**
- Randomness is the key — not just complexity

Length + Randomness + Mixed Characters = Strong Security

5. Common Password Attack Techniques

Attack Type	Description	Targets Weakness
Brute Force	Tries every possible combination	Short, simple passwords
Dictionary Attack	Uses list of common words/names	Passwords using names / dictionary words
Credential Stuffing	Reuses breached passwords	Same password used across accounts
Phishing	Trick user into revealing password	Human error
Keylogging	Malware records keystrokes	Unprotected devices
Shoulder Surfing	Observing you typing password	Public environments

6. Why Weak Passwords Fail

Weak passwords are:

- **Often already leaked** in breach databases
- Based on **personal details** (easy to guess)
- Very short → **low brute force complexity**

- Predictable patterns like:
 - Password123
 - Name@123
 - 123456 / qwerty

7.Password Best Practices — Based on NIST Security Standards

- ✓ Use **minimum 12–16 characters**
- ✓ Use **uppercase + lowercase + numbers + symbols**
- ✓ Prefer **random passphrases** instead of names or quotes
- ✓ Avoid using **personal information**
- ✓ Use **different password** for every account
- ✓ Store passwords in a **password manager**
- ✓ Enable **Multi-Factor Authentication (MFA)**
- ✓ Change password **only after suspicious activity** (NIST guideline)

Recommended Passphrase Format

Unrelated Words + Symbols + Numbers

Example: Tiger!Sky7\$LaptopGreen

8. Conclusion

This task demonstrated that password strength greatly depends on:

Security Factor Impact

Length More characters = mathematically harder to crack

Complexity Mixed characters disrupt pattern attack

Unpredictability Prevents dictionary and brute-force success

Passphrases Best combination of strength + memorability

Final Result: **Strong passwords can take centuries to crack**, while weak ones can fall in seconds.
By applying password security best practices and using MFA/password managers, overall cybersecurity posture significantly improves.